Concepts of Biostatistics and Data Science

A two-day crash course with practical analysis in R: overview, timetable and pre-course preparation

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About the crash course

Concepts of Biostatistics and Data Science is a two-day crash course for teaching the core ideas of biostatistics and the first steps of data science. It is designed to be delivered as a short, intensive course: each day has two hours of explanation and two hours of guided analysis in R. The emphasis is on choosing a method for a stated question, checking its assumptions and interpreting the result in plain language. R commands are taught as the means to that end.

As a crash course it covers the essentials and points to further learning rather than trying to teach every method in depth. Trainers can use the slides, workbook, scripts and reference pack as a complete package, or take individual sessions for shorter teaching.

Day 1 covers understanding data and making comparisons: research questions, variables and data quality, descriptive statistics and plots, sampling variability, confidence intervals and tests, comparisons of two or more groups, and associations between categorical variables. Day 2 moves from relationships between variables to simple and multiple linear regression, confounding and mediation, model checking, and short introductions to logistic regression, Poisson regression, prediction and clustering.

> All examples use one simulated dataset, the Healthy Hearts District Study. It was generated with known rules so that results can be checked. Any finding from it applies only to the simulation and is for teaching.

Who it is for

- People with little statistical training who need to understand or carry out basic analyses
- Data enthusiasts who want a sound statistical footing for working with data
- Participants with some working knowledge who want to strengthen their understanding and practical skills. Optional extension exercises are provided for them.

Prerequisites

- Comfort with basic arithmetic, percentages and reading a simple graph
- No previous statistics course is required
- No previous R experience is required, but completing the pre-course preparation (below) is essential
- A laptop on which you can install software (Windows, macOS or Linux)

Learning outcomes

By the end of the course participants will be able to:

#	Outcome	Day	How it is practised
1	Identify observations, variables and variable types in a dataset, and state a research question in terms of an outcome, exposure or groups, and design	1	Workbook 1.1 to 1.3
2	Import a CSV file into R, check for missing and implausible values, and recode a missing-value code	1	Workbook 1.1 to 1.2
3	Calculate and report counts with denominators, proportions, mean (SD) and median (IQR), and justify the choice	1	Workbook 1.4 to 1.5
4	Produce and describe a bar chart, histogram and boxplot	1	Workbook 1.6
5	Interpret a 95% confidence interval and a p-value correctly, and distinguish statistical significance from practical importance	1	Workbook 1.7 to 1.9
6	Compare two or more independent groups with Welch's t-test, ANOVA, Wilcoxon rank-sum or Kruskal-Wallis tests, and explain what each compares	1	Workbook 1.7, 1.9, E1.1
7	Recognise paired data and analyse it with a paired method	1	Workbook 1.8, E1.4
8	Analyse a contingency table: proportions with explicit denominators, risk or prevalence ratio, odds ratio, chi-square and Fisher's exact tests	1	Workbook 1.10 to 1.11
9	Describe a relationship with a scatterplot and correlation, and explain why association is not causation	2	Workbook 2.1
10	Fit, interpret and check simple and multiple linear regression models, including adjusted associations	2	Workbook 2.2 to 2.4
11	Explain confounding and mediation with a causal diagram, and decide what to adjust for	2	Workbook 2.3
12	Read and interpret the output of a logistic regression (OR, aOR) and a Poisson regression with an offset (IRR), including a check for overdispersion	2	Workbook 2.5 to 2.6 (guided)
13	Explain the difference between describing, estimating associations, predicting and discovering groups, including train/test separation and data leakage	2	Slides; extension E2.2
14	Run and interpret a k-means clustering with scaled variables, and explain why clusters need checking	2	Workbook 2.7 (guided)
15	Choose an analysis for a stated question, produce a suitable plot and write a plain-language conclusion	2	Final exercise

What is practised, demonstrated or left for further reading

Eight hours is enough to build sound foundations, not to master every method. The table sets realistic expectations.

Level	Topics
Practised in R (core pathway)	Importing and checking data; descriptive statistics and plots; Welch t-test; paired t-test; one-way ANOVA with pairwise comparisons; chi-square and Fisher's exact tests; risks, odds, RR and OR; correlation; simple and multiple linear regression; residual checks; plotting fitted values
Guided demonstration (run the code, interpret the output)	Wilcoxon and Kruskal-Wallis tests; Mantel-Haenszel adjustment; logistic regression and aOR; Poisson regression with offset and overdispersion; k-means clustering
Short demonstration or extension	McNemar's test; ANCOVA for a trial; train/test split and RMSE; negative binomial regression; hierarchical clustering; interaction terms
Further learning (signposted only)	Ordinal and multinomial regression; survival analysis; mixed models and GEE for repeated or clustered data; multiple imputation; formal causal inference; machine learning beyond the basics

Timetable

Each day has 240 minutes of teaching: 120 minutes of theory and 120 minutes of practical work. The break is listed separately and is not counted in teaching time. Clock times are an example and can be shifted.

Day 1: Understanding data and making comparisons

Time	Minutes	Session	Content	Materials
09:00	10	Theory	Welcome, aims, the course dataset, how the days work	Day 1 slides 1 to 4
09:10	20	Theory	Questions, observations, variables and types; data quality and missing values	Slides, Part 1
09:30	25	Theory	Counts and proportions; mean, SD, median, IQR; skewness; bar charts, histograms, boxplots	Slides, Part 2
09:55	25	Theory	Samples and populations; variability; SE and CIs; effect sizes; p-values; significance vs importance	Slides, Part 3
10:20	22	Theory	Independent vs paired; Welch t-test; skewed outcomes; paired t-test; ANOVA, pairwise comparisons, Kruskal-Wallis	Slides, Part 4
10:42	18	Theory	Contingency tables; risks, odds, RR, OR; design and measures; chi-square, Fisher; first look at confounding; McNemar	Slides, Part 5
11:00	(30)	Break	Not counted in teaching time	
11:30	25	Practical A	Import, inspect, recode -99, missing values, factor levels	01_import_and_inspect.R; workbook 1.1 to 1.3
11:55	30	Practical B	Frequencies, summaries, histograms, boxplots	02_describe_and_plot.R; 1.4 to 1.6
12:25	35	Practical C1	t-tests, log scale, Wilcoxon, paired t-test, ANOVA	03_compare_groups.R; 1.7 to 1.9
13:00	20	Practical C2	2 x 2 tables, RR and OR, chi-square, Fisher	04_categorical.R; 1.10 to 1.11
13:20	10	Review	Questions, common mistakes, preview of Day 2	
	240	Total	120 theory + 120 practical	

Day 2: Regression and an introduction to data science

Time	Minutes	Session	Content	Materials
09:00	10	Theory	Recap; describing, estimating associations, predicting, discovering groups	Day 2 slides 1 to 6
09:10	15	Theory	Scatterplots; Pearson and Spearman correlation; association and causation	Slides, Part 1
09:25	20	Theory	Simple linear regression: the line, output, assumptions, residuals, categorical predictors	Slides, Part 2
09:45	25	Theory	Multiple regression; confounders and mediators; choosing adjustment variables; interpreting coefficients	Slides, Part 3
10:10	10	Theory	Plotting fitted relationships; effect modification	Slides, Part 3
10:20	20	Theory	Logistic regression (OR, aOR, probabilities); Poisson regression (rates, offsets, overdispersion)	Slides, Part 4
10:40	20	Theory	Explaining vs predicting; train/test and leakage; k-means, scaling, choosing k, hierarchical clustering; further learning	Slides, Part 5

Time	Minutes	Session	Content	Materials
11:00	(30)	Break	Not counted in teaching time	
11:30	50	Practical A	Correlation, simple and multiple regression, confounding, model checks, fitted plots	05_correlation_regression.R; 2.1 to 2.4
12:20	20	Practical B	Guided: logistic and Poisson regression	06_logistic_poisson.R; 2.5 to 2.6
12:40	20	Practical C	Guided: k-means clustering	07_clustering.R; 2.7
13:00	25	Final exercise	Choose an analysis, plot, conclusion	08_final_exercise.R
13:25	5	Close	Reporting checklist; resources for further learning	
	240	Total	120 theory + 120 practical	

If the group is slower than planned, shorten Day 1 Part 5 (the practical revisits it) and turn Day 2 Practical C into a five-minute demonstration. Do not cut the final exercise.

Pre-course preparation (about 30 minutes)

Please complete these steps before Day 1. Installation problems are the most common way to lose teaching time.

1. Install R (version 4.1 or later) from CRAN: https://cran.r-project.org/ . Choose the download for your operating system and accept the default options. R is the program that does the calculations.
2. Install RStudio Desktop (free, open source) from Posit: https://docs.posit.co/ide/user/download.html . RStudio is the window you work in: it shows your script, the console, plots and data side by side. Install R first.
3. Unzip the course folder to a location you can find, for example Documents.
4. Double-click `Concepts-of-Biostatistics-and-Data-Science.Rproj`. RStudio opens with the course folder as the working folder, so paths such as `data/hearts.csv` work.
5. In RStudio, open `R/00_install_and_check.R` and click Source. It installs the four course packages (readr, dplyr, ggplot2, broom) and checks that the data can be read.
6. Check that the last line in the console says: Your computer is ready for the course.

Links checked on 26 October 2026. Software websites change; if a link has moved, search for "CRAN R download" or "RStudio Desktop download".

If installation fails

- Bring the laptop anyway: the instructor will help at the start of Day 1, or you can pair with a neighbour.
- The cleaned data are supplied ready to use as `data/hearts.csv` and `data/hearts.rds`, so no download is needed during the course.
- If packages cannot be installed, all tests and models still run in base R. Use `read.csv()` instead of `read_csv()`; ggplot2 plots can be replaced by `hist()`, `boxplot()` and `plot()`.
- If the data files are missing or damaged, `source("data/make_hearts_data.R")` rebuilds them using base R only.
- As a last resort, Posit Cloud (https://posit.cloud) runs RStudio in a web browser. Check its current account options before the course.

R and RStudio in one paragraph

R is a free programming language for statistics. You give it instructions (commands) and it returns results, tables and plots. Commands are saved in a script (a text file ending in .R) so that the whole analysis can be repeated exactly. RStudio is a free program that makes R easier to use: the script is in the top-left pane, the console where results appear is below it, the data and objects you have created are listed top-right, and plots and help appear bottom-right. Run the current line with Ctrl+Enter (Cmd+Enter on a Mac). Add-on packages extend R; we use four of them.

Crash course materials

Material	File(s)
Slides with speaker notes	slides/Day1_*.pptx, slides/Day2_*.pptx (and PDF copies)
Participant workbook (includes cheat sheet and decision diagrams)	workbook/Participant_workbook.docx (and PDF)
Final exercise	workbook/Final_exercise.docx; R/08_final_exercise.R
R scripts in teaching order	R/00 to R/08
Data and data dictionary	data/hearts_raw.csv, data/hearts.csv, data/hearts.rds, data/data_dictionary.csv
Cheat sheet, decision diagrams, method-selection guide	reference/ (PDF and editable LaTeX sources)
Instructor answer key and solutions	instructor/ (answer key, solutions.R, output logs)